# Theme and chassis layout



**antocciphysics | TEACHER KEY | CPA + MATCHING SEI**

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**Learning target:** Make a dimensioned layout that supports the circuit and drivetrain.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 7, Day 18.*
24 J; heating or sound are possible.

**02. Worked example: explain the model**
Length account 20=6+5+2+7 cm; real build still needs measured component widths/heights/clearances.

**03. Read and interpret the model**
Student-specific measured layout with feasible spacing and accessible controls.

**04. Apply the physics**
18-6-5=7 cm; actual plan must include gaps and component dimensions.

**05. Explain your reasoning**
May alter balance, wheel loading or clearance; compare matched runs or inspect/measure loading and straight tracking.

**06. Review: connect an earlier idea** *Review: Unit 7, Day 9.*
8 A in each branch.

**07. Exit check**
Student supplies actual critical dimension and approval status; no automatic approval.

**08. STRETCH: test the model**
Remaining 5 cm after stated lengths/clearances; layouts must avoid overlap and allow wiring/axles; measure actual dimensions.

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Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 7, Day 18.*
$|emf| = N|\Delta\Phi|/\Delta t = 0.6$ V.

**02. Worked example: explain the model**
Length account $20 = 6+5+2+7$ cm; real build still needs measured component widths/heights/clearances.

**03. Read and interpret the model**
Student-specific measured layout with feasible spacing and accessible controls.

**04. Apply the physics**
55 mm remains; 0.180 m.

**05. Explain your reasoning**
May alter balance, wheel loading or clearance; compare matched runs or inspect/measure loading and straight tracking.

**06. Review: connect an earlier idea** *Review: Unit 7, Day 9.*
Parallel 6 Ω; total 14 Ω; supply current 2 A.

**07. Exit check**
Height/thickness interference may be hidden; include axle height, gear clearance or battery height.

**08. STRETCH: test the model**
40 mm remains. Gear radius, axle height, thickness or battery height may constrain side view.